ESKWELA Software Testing & Quality Assurance (QA) Suite

Project: ESKWELA Android Mobile Application | **QA Lead:** Software Testing & QA Engineer

Code Repository Link: https://github.com/jameshofril06-jpg/SkwelaApp

Postman Collection Link: Documentation/Pre-Fi/eskwela_postman_collection.json

Total Executed Test Cases: 30 Sample Test Cases (18 Functional, 8 Integration/API, 4 UAT Scripts)

1. Executive Test Plan & QA Strategy

This Pre-Final QA document outlines the software quality assurance test suite for **ESKWELA**, containing 30 documented test cases, integration API test scripts, User Acceptance Testing (UAT) scenarios, and an initial bug tracking resolution matrix. Testing verifies functional correctness across Student and Administrator workflows.

2. Functional Testing Suite (18 Test Cases)

Functional Test Suite (TC-01 to TC-18)

TC ID	Module	Test Condition / Step Description	Expected Verification Result	Status
TC-01	Registration	Register with valid student credentials and password	Account created, redirected to Profile Setup	PASSED
TC-02	Login	Enter valid registered student credentials	Authenticated successfully, opens Home Dashboard	PASSED
TC-03	Login	Tap 'Forgot Password' link and submit email	Password reset email dispatched via Firebase	PASSED
TC-04	Profile Setup	Enter Full Name, School, Course, Year, Section	Profile stored in Firestore document	PASSED
TC-05	Profile Picture	Tap avatar circle to pick custom profile picture	Image selected from gallery and saved to profile	PASSED
TC-06	Password Toggle	Tap eye icon on password input field	Toggles visibility between hidden dots and plain text	PASSED
TC-07	Account Guard	Attempt login with suspended student status	Blocked with restriction notice, session terminated	PASSED
TC-08	Admin Auth	Login using administrator email credentials	AccountGuard redirects directly to Admin Center	PASSED
TC-09	Admin Isolation	Admin attempts opening student schedule tab	Access restricted, redirects to Admin Center (Home & Profile only)	PASSED
TC-10	Logout	Tap Logout button in Profile settings	Displays confirmation dialog, revokes session upon confirmation	PASSED
TC-11	Dark Mode	Toggle Dark Theme switch in Profile settings	App appearance instantly flips to dark theme with high-contrast text	PASSED
TC-12	OTA Check	Tap 'Check for update (OTA)' in Profile settings	Queries GitHub releases API and displays release notes	PASSED
TC-13	OCR Timetable	Scan study load document via Camera or Gallery	ML Kit extracts courses and plots across weekdays	PASSED
TC-14	Day View	Tap specific day tab on schedule calendar	Displays strictly classes scheduled for that specific date	PASSED
TC-15	Time Conflict	Add class overlapping existing class time	Displays ■■ Conflict warning with override option	PASSED
TC-16	Class Deletion	Long press class card and type matching Class Code	Validates code and deletes class from schedule	PASSED
TC-17	Task Management	Create task with priority, due date, and course link	Task saved, rendered in task list with priority badge	PASSED
TC-18	AI Quiz Generator	Select study load course and complete practice quiz	Generates course questions, tracks score, shows explanations	PASSED

3. Integration & REST API Test Suite (8 Test Cases)

Integration & REST API Testing (INT-01 to INT-08)

TC ID	Module	Test Condition / Step Description	Expected Verification Result	Status
INT-01	Firebase Auth API	POST /v1/accounts:signUp with email/pass	Returns HTTP 200 OK with idToken and refreshToken	**PASSED**
INT-02	Firebase Auth API	POST /v1/accounts:signInWithPassword	Returns HTTP 200 OK with authenticated user UID	**PASSED**
INT-03	Firestore Users API	GET /documents/users/{userId}	Returns user profile document JSON fields	**PASSED**
INT-04	Firestore Users API	PATCH /documents/users/{userId} status='suspended'	Updates user status and suspensionDuration in Firestore	**PASSED**
INT-05	Firestore Schedules	POST /documents:runQuery for schedules dateString	Returns JSON array of enrolled classes for date	**PASSED**
INT-06	Firestore Tasks API	POST /documents/tasks (Add Task)	Creates task document with priority and course link	**PASSED**
INT-07	Firestore Quiz API	POST /documents/ai_quizzes (Save Quiz Attempt)	Saves score, topic, percentage, timestamp	**PASSED**
INT-08	GitHub Releases API	GET api.github.com/repos/.../releases	Returns HTTP 200 OK JSON array of release tags and assets	**PASSED**

4. User Acceptance Testing (UAT) Scripts (4 Test Cases)

User Acceptance Testing (UAT) Scenario Scripts (UAT-01 to UAT-04)

TC ID	Module	Test Condition / Step Description	Expected Verification Result	Status
UAT-01	Student Onboarding	Register ➔ Setup Profile & Avatar ➔ Scan Timetable	Student successfully completes onboarding flow	**PASSED**
UAT-02	Study Load & Schedule	View Home Dashboard (Today's Classes) ➔ Switch Day Tabs	Dashboard and schedule tabs filter classes by specific date	**PASSED**
UAT-03	AI Quiz & Leaderboard	Take course-specific AI Quiz ➔ View Leaderboard rankings	Quiz completed, score saved, and leaderboard rendered	**PASSED**
UAT-04	Admin Moderation	Admin logs in ➔ Inspect user ➔ Suspend account ➔ Broadcast announcement	Admin moderation executed and broadcasted successfully	**PASSED**

5. Initial Bug Tracking Log & Resolution Matrix

Bug ID	Description / Defect	Severity	Root Cause Analysis	Resolution / Fix Applied	Status
BUG-01	Scanned study load did not reflect on specific day tabs	HIGH	Schedules were only seeded for specific dates instead of full weekdays.	Updated ScheduleActivity to plot study load across specific weekdays.	RESOLVED
BUG-02	OTA update download failed with HTTP 404	HIGH	DownloadManager failed on GitHub's /releases/latest/ 302 redirect.	Switched to GitHub /releases API index 0 and added cache-busting timestamp.	RESOLVED
BUG-03	Text unreadable when editing profile in Dark Mode	MEDIUM	Profile setup layout contained hardcoded dark hex text colors.	Replaced hardcoded hex colors with dynamic @color/text_primary tokens.	RESOLVED
BUG-04	Time schedule overlap allowed duplicate class booking	MEDIUM	AddClassActivity lacked start/end time overlap validation.	Implemented parseTimeToMinutes() and overlap check with warning popup.	RESOLVED
BUG-05	Admin user could access student schedule/tasks tabs	HIGH	Role access guard lacked strict navigation redirection for admin users.	Implemented AccountGuard.redirectIfAdmin() routing admins exclusively to Admin Center.	RESOLVED